

DATABASE MANAGEMENT SYSTEMS

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ERR MODEL

(Enhanced Entity Relationship Model)

Extends the Basic ER Model to represent more complex real-world scenarios

1. Generalization

Bottom-up approach
Combines similar entity sets into a higher-level supertype

Example

Person

Student Employee Teacher

Meaning

STUDENT, EMPLOYEE and TEACHER are specialized kinds of PERSON.

Common attributes are stored in PERSON

2. Specialization

Top-down approach
Divides a higher-level entity set into lower-level specialized entity sets

Example

Vehicle

Car Truck Bike

Meaning

VEHICLE is specialized into CAR, TRUCK and BIKE based on distinguishing characteristics

3. Aggregation

Treats a relationship set as a higher-level entity so that it can participate in another relationship

Example

Student Enrolls Course

ASSESSED_BY

FACULTY

Meaning

The relationship ENROLLS between STUDENT and COURSE is treated as a higher-level entity (aggregation) so it can be related to FACULTY via ASSESSED_BY.

BENEFITS OF EER MODEL

- * More expressive - captures more real-world constraints
- * Reduces redundancy by using inheritance
- * Supports modelling of complex relationship (aggregation)
- * Produces a better conceptual design for databases.

(2)

